Low-Distortion Dual-Cell Power Amplifier with Ultra-High Small Signal Bandwidth

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***Abstract*—This paper presents a dual-cell hybrid power amplifier with an ultra-high small-signal bandwidth. A control strategy is introduced that maximizes the achievable bandwidth while decoupling the operation of the power-amplifier cells. To validate the concept, a two-phase full-bridge prototype is implemented, with the main switching devices operating at 200 kHz and the auxiliary switching devices at 4 MHz. Step-response tests show an output current rise-time of about 1.5 µs and a settling time of roughly 90 µs in closed-loop operation. When tracking a 64 Hz, 24 A-peak sinusoidal current reference, the amplifier achieves a total harmonic distortion of approximately -100 dB across the first 150 harmonics. It is demonstrated that the approach enables the combination of about 6.45 kW output power and an 8 MHz effective switching frequency while maintaining high conversion efficiency.**

***Index Terms*—High-precision power amplifiers, high bandwidth, closed-loop control, THD.**

I. Introduction

Power amplifiers are essential in many applications. For example, in high-precision motion systems, they drive linear actuators to produce displacements with sub-nanometer accuracy. To meet this requirement, a power amplifier must convert a digital reference signal into a precise output current that generates the actuation force needed to achieve the desired movement [1], [2]. High-precision output currents are achieved mainly through output waveform closed-loop and feedforward control strategies [3], [4].

Recent developments have explored advanced control and topology techniques to enhance dynamic performance. Tang et al. proposed a Low-Carrier-Ratio Model Predictive Control (LCR-MPC) method that enables wide-bandwidth operation with reduced Total Harmonic Distortion (THD) but suffers from implementation complexity and limited scalability [5]. Cui et al. proposed an Auxiliary Commutated Resonant Pole (ACRP) converter that achieves ultra-low distortion and full-range soft switching but suffers from high conduction losses due to increased current stress on the transistors and auxiliary inductors [6]. Niklaus et al. developed a high-power GaN-based Class-D amplifier that employs series–parallel interleaving to achieve a 100 kHz large-signal bandwidth with high efficiency; however, its hardware is overdesigned for applications where only small-signal bandwidth improvement is required [7].

In the high-precision motion systems studied in this research, the current reference is relatively slow and, by design, the majority of its power lies in low-frequency components. However, to achieve high positioning accuracy, the power amplifier must track the high-frequency components and quickly reject any unwanted disturbances. Therefore, a high small-signal bandwidth is sufficient for accurate reference tracking [3].

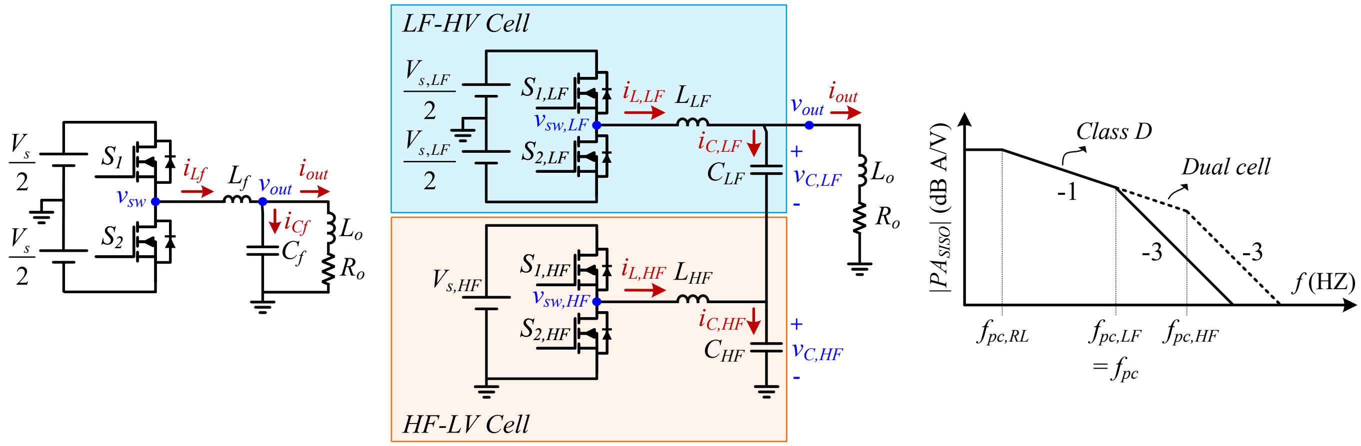
To achieve a high small-signal bandwidth, the analog/digital hybrid amplifiers introduced by Yundt [8] are of particular interest. The analog stage can be replaced with a switched-mode (digital) amplifier which, because of its lower power requirements, can operate at much higher switching frequencies than the main amplifier [9].

A digital/digital hybrid amplifier that connects the main and auxiliary cells in parallel and uses capacitive coupling between them is a promising topology for improving the accuracy of power amplifiers. In this paper, we refer to this topology as a dual-cell power amplifier because it comprises two cells: a main Low-Frequency High-Voltage (LF-HV) cell, which delivers the bulk of the output power, and a correction High-Frequency Low-Voltage (HF-LV) cell, which extends the small-signal bandwidth and enables fine control of the output power. By contrast, the conventional class-D power amplifier is referred to as a single-cell converter [3], [10].

The dual-cell power amplifier was introduced and analyzed in [3]. In this paper, we present a new control strategy that increases the achievable bandwidth of the dual-cell power amplifier and decouples the control actions in the frequency domain by using digital high-pass (and optionally low-pass) filters. Furthermore, a 400 V/24 A prototype has been designed and implemented to verify the performance of the proposed dual-cell power amplifier using a resistive 11.2 Ω load.

II. Proposed dual-cell power amplifier

Fig. 1(a) and (b) depict the single-phase half-bridge representations of a conventional single-cell and the proposed dual-cell power amplifiers, respectively. Compared with the single-cell design, the dual-cell amplifier introduces a correction-cell output port connected in series with the main-cell output capacitor to improve the control bandwidth of the system. Fig. 1(c) shows the plant of the proposed dual-cell power amplifier in comparison with that of the single-cell amplifier.



(a) (b) (c)

Fig. 1. One-phase half-bridge representation of (a) conventional single-cell, and (b) proposed dual-cell power amplifiers. (c) Conceptual PA_{SISO} gain of proposed dual-cell power amplifier compared with that of single-cell power amplifier.

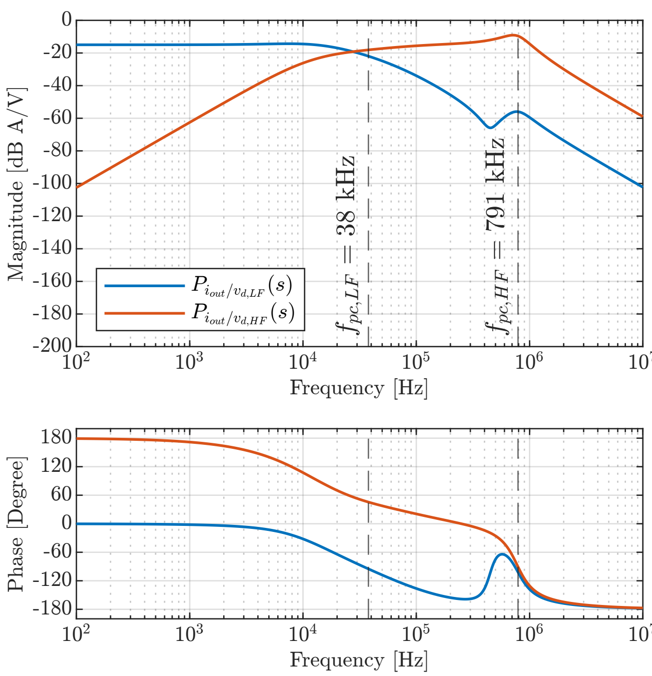


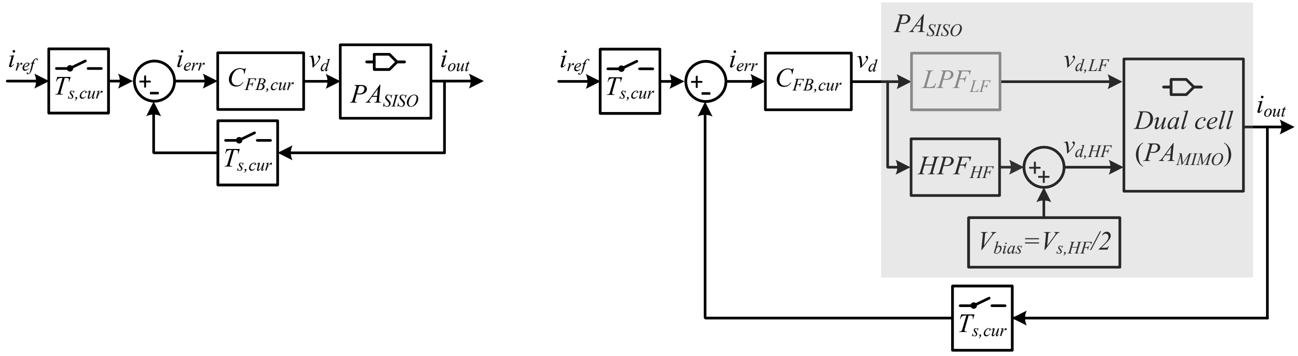
Fig. 2. Bode plots of $PA_{MISO}(s)$ with transfer functions from first and second inputs to output, $P_{i_{out}/v_{d,LF}}(s)$ and $P_{i_{out}/v_{d,HF}}(s)$.

As no additional components are introduced in the main power path, the losses and the voltage and current stress on the HF-LV cell remain minimal, leading to negligible efficiency and cost penalties. Moreover, a unipolar voltage source is sufficient for powering the HF-LV cell. Unlike the converter in [4], no isolated voltage source is used. Hence, there is no need for transformers nor power-factor-correction converters to construct the HF-LV cell voltage source. The main challenge of implementing the proposed dual-cell converter is to control both the LF-HV and HF-LV cells such that the control bandwidth is increased while the whole system remains stable.

III. Controller design

In this section, the design of a current controller for a purely resistive load is explained. RC passive damping circuits are connected in parallel with C_{LF} and C_{HF}, thereby eliminating the need for active damping control. The proposed converter State Space Average (SSA) model represents a Multi-Input Single-Output (MISO) system with two inputs and one output. The inputs are defined as $u = [v_{d,LF} \quad v_{d,HF}]^T$, representing the average switching-node voltages over one switching cycle (i.e. $v_{d,LF} = \overline{v_{sw,LF}}$ and $v_{d,HF} = \overline{v_{sw,HF}}$). The output and state variable vectors are defined as $y = [i_{out}]$ and $x = [i_{L,LF} \quad i_{L,HF} \quad v_{C,LF} \quad v_{C,HF} \quad v_{Cd,LF} \quad v_{Cd,HF}]^T$, respectively. The SSA matrices are presented in Appendix A. Fig. 2 shows the Bode plot of this MISO system, illustrating the transfer functions from the inputs to the output.

In our previous work [3], we proposed a control strategy with three control loops—two inner voltage-control loops and one outer current-control loop. However, this cascaded structure both limits the achievable control bandwidth and requires additional voltage measurements. This paper introduces a modified control strategy that increases the bandwidth of the dual-cell power amplifier. Fig. 3 illustrates the proposed control strategy alongside the control diagram of the single-cell power amplifier. In this scheme, the inner controllers are replaced by frequency-selective filters—a high-pass filter (HPF_{HF}) and, optionally, a low-pass filter (LPF_{LF})—to decouple the control actions of the HF-LV and LF-HV cells in the frequency domain, effectively transforming PA_{MISO} into a Single-Input Single-Output (SISO) plant, PA_{SISO}. The input vector of $PA_{SISO}(s)$ is $u = [v_d]$, and the B and D



(a) (b)

Fig. 3. Control diagram of (a) conventional single-cell, and (b) proposed dual-cell power amplifiers.

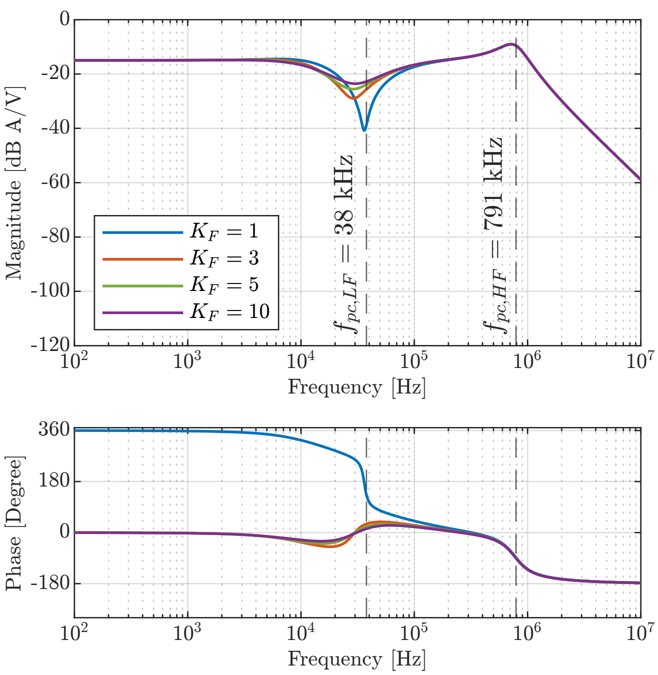


Fig. 4. Bode plot of $PA_{SISO}(s)$ for different values of K_F.

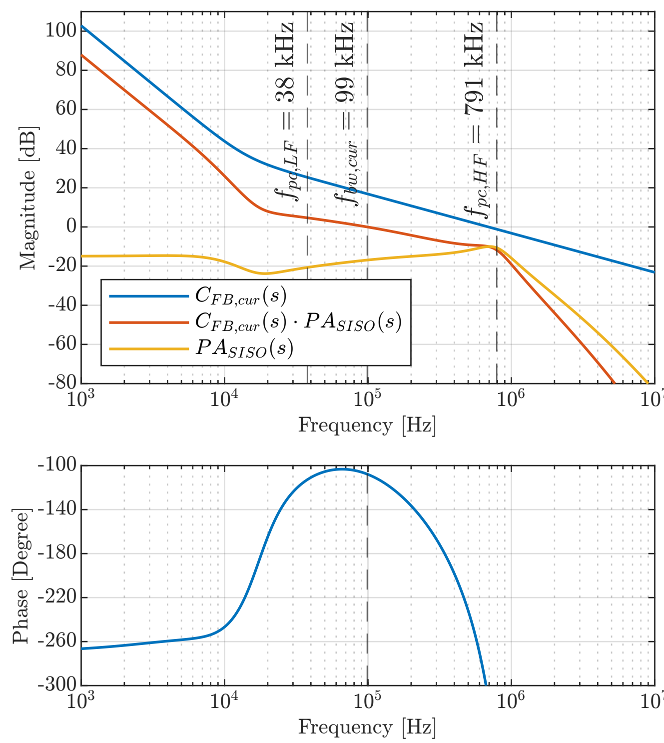


Fig. 5. Bode plot of the designed $C_{FB,cur}(s)$ used to shape the OL response $C_{FB,cur}(s) \cdot PA_{SISO}(s)$ toward a (-3, -1, -3) slope characteristic.

matrices of the SSA model are modified as follows:

$$B = [LPF_{LF}(s)/L_{LF} \quad HPF_{HF}(s)/L_{HF} \quad 0 \quad 0 \quad 0 \quad 0]^T,$$
$$D = [0]. \tag{1}$$

Additionally, to avoid the need for a bipolar voltage source for the HF-LV cell, a bias voltage $V_{bias} = V_{s,HF}/2$ is added to its input.

The continuous-time transfer functions of the LPF_{LF} and HPF_{HF} blocks are defined as

$$LPF_{LF}(s) = \frac{\omega_{LF}}{s + \omega_{LF}}, \quad HPF_{HF}(s) = \frac{s}{s + \omega_{HF}}, \tag{2}$$

where ω_{LF} and ω_{HF} are the cutoff frequencies for $LPF_{LF}(s)$ and $LPF_{LF}(s)$, respectively, and are given by

$$\omega_{LF} = 2\pi f_{LF}, \text{ where } f_{LF} = f_{pc,LF} \cdot K_F,$$
$$\omega_{HF} = 2\pi f_{HF}, \text{ where } f_{HF} = f_{pc,LF}/K_F. \tag{3}$$

Here, K_F is a constant that defines the overlap between the HF-LV and LF-HV cell frequency responses, and $f_{pc,LF}$ is the cut-off frequency of the LF-HV cell. Fig. 4 shows how the shaped SISO plant's gain and phase vary with different values of K_F. As K_F increases, the overlap between the HF-LV and LF-HV cell frequency responses also increases, resulting in lower gain and phase fluctuations around $f_{pc,LF}$. On the other hand, a larger K_F, also means a higher HF-LV cell controller effort is required to compensate for disturbances in the overlapping frequency range. Therefore, $K_F = 5$ is selected because it keeps the gain and phase fluctuations below 5% and 10%, respectively. For $K_F = 5$, the cutoff frequency of LPF_{LF} is $5f_{pc,LF}$, which lies within the LF-HV cell's switching frequency range. As a result, the LPF_{LF} would be ineffective and can therefore be omitted from the design.

Fig. 5 shows the Bode plot of the designed $C_{FB,cur}(s)$, which shapes the Open-Loop (OL) response of $C_{FB,cur}(s) \cdot PA_{SISO}(s)$ to achieve a (-3, -1, -3) slope profile. The general form of $C_{FB,cur}(s)$ is given by:

$$C_{FB,cur}(s) = k_{dc} \cdot \frac{s^2 + 2\zeta_{zc}\omega_{zc}s + \omega_{zc}^2}{s^3}. \tag{4}$$

After discretization with $T_{s,cur}$, $C_{FB,cur}(z)$ becomes:

$$C_{FB,cur}(z) = \frac{v_d(z)}{i_{err}(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3}}. \tag{5}$$

where the denominator coefficients are:

$$a_0 = 1, \quad a_1 = -3, \quad a_2 = 3, \quad a_3 = -1. \tag{6}$$

The numerator coefficients depend on the discretization method used. For the direct discretization method (detailed in [11], [12]), they are computed as:

$$b_0 = \frac{T_{s,cur}^3 k_{dc}\omega_{zc}^2}{2(1 - 2p_z\cos(q_z) + p_z^2)}, \quad b_1 = b_0 \cdot (1 - 2p_z\cos(q_z)), \tag{7}$$

$$b_2 = b_0 \cdot (p_z^2 - 2p_z\cos(q_z)), \quad b_3 = b_0 \cdot p_z^2,$$

with

$$p_z = \exp(-\omega_{zc}T_{s,cur}\zeta_{zc}), \quad q_z = \omega_{zc}T_{s,cur}\sqrt{1 - \zeta_{zc}^2}. \tag{8}$$

The main controller is implemented as a third-order Infinite Impulse Response (IIR) filter, as expressed in

$$v_d[n] = \sum_{k=0}^{3} b_k\, i_{err}[n-k] - \sum_{k=1}^{3} a_k\, v_d[n-k]. \tag{9}$$

To achieve the required 4-MHz update rate required for this power amplifier, the controller the controller is realized on an Artix-7 FPGA using dedicated DSP slices operating in parallel.

IV. Experimental verification

To validate the performance of the proposed dual-cell power amplifier, we have implemented a two-phase interleaved full-bridge prototype in the laboratory and tested it with a resistive load. Table I lists the power amplifier's specifications. Fig. 6 shows the implemented prototype, which consists of multiple custom-designed PCBs: an HF-LV switching-legs board, an LF-HV switching-legs board, an output-filter board, an FPGA SoM and interface board, and dedicated boards for current and voltage measurement. The measurement results were recorded and compared with the simulation results (SIM). Data were acquired in two ways: (1) with an oscilloscope (OSC) using current and voltage probes, and (2) with the FPGA's Integrated Logic Analyzer (ILA) core, which sampled the ADC-measured data.

TABLE I
Prototype characteristics.

Input parameter	Symbol	Value	Unit
LF-HV cell supply voltage	$V_{s,LF}$	400	V
HF-LV cell supply voltage	$V_{s,HF}$	24	V
LF-HV cell switching frequency	$f_{sw,LF}$	200	kHz
HF-LV cell switching frequency	$f_{sw,LF}$	4	MHz
LF-HV cell filter inductor	L_{LF}	59	µH
LF-HV cell filter capacitor	C_{LF}	600	nF
HF-LV cell filter inductor	L_{HF}	270	nH
HF-LV cell filter capacitor	C_{HF}	300	nF
LF-HV cell passive damping capacitor	$C_{d,LF}$	5	µF
LF-HV cell passive damping resistor	$R_{d,LF}$	4	Ω
HF-LV cell passive damping capacitor	$C_{d,HF}$	2	µF
HF-LV cell passive damping resistor	$R_{d,HF}$	1.6	Ω
Peak output current	$I_{out_{\max}}$	24	A
Load resistance	R_o	11.2	Ω

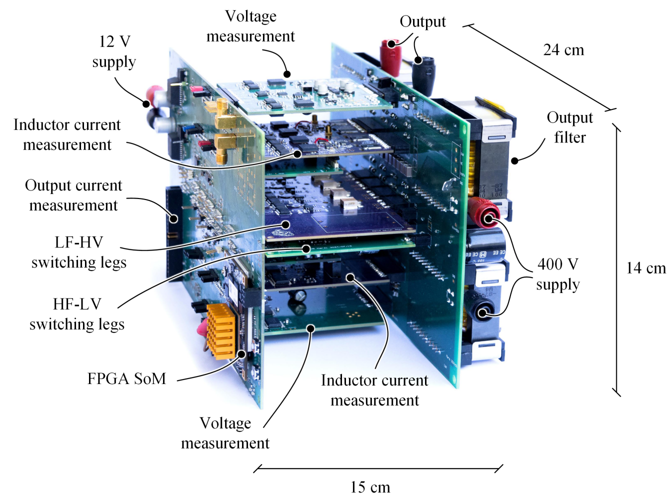
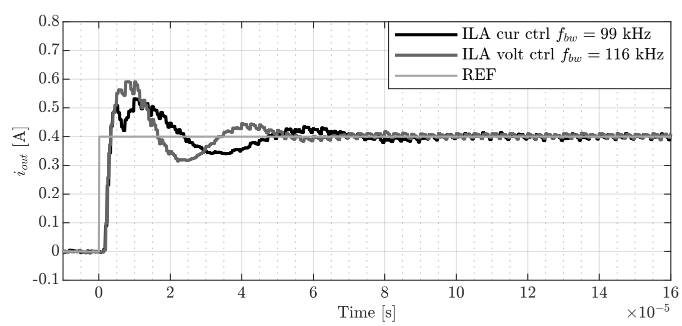


Fig. 6. Assembled prototype side view.



(a)

Fig. 7. Step current comparison between voltage and current controllers.

Fig. 7 shows the step response of the power amplifier in two modes: output-current control and output-voltage control. For a fair comparison, the reference voltage is chosen to generate an equivalent 0.4 A current step. It should be noted that the proposed dual-cell power amplifier is restricted to small-signal

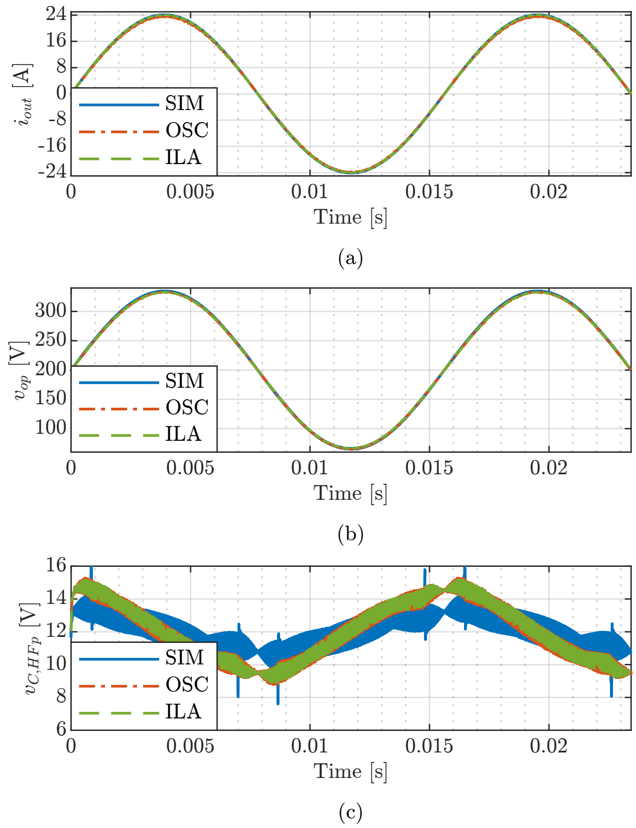


Fig. 8. Sinusoidal current reference. The sampling frequencies for SIM, OSC, and ILA are 2 MSPS, 31.25 MSPS, and 1 MSPS, respectively.

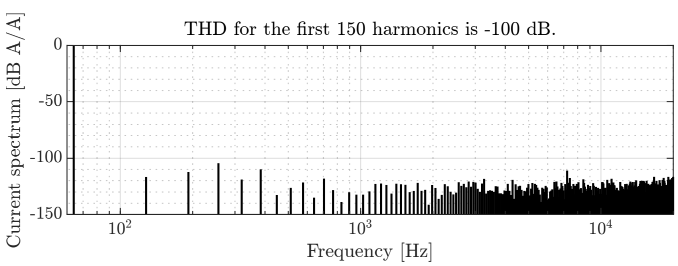


Fig. 9. Normalized spectrum of the output current, derived from 1 MSPS ILA data for a 24 A, 64 Hz sinusoidal reference.

steps owing to the saturation limit in the HF-LV cell. As the figure shows, the output-voltage controller settles about 20 µs faster but exhibits roughly 20% more overshoot than the output-current controller. This occurs because the voltage-measurement path employs faster 4 MSPS ADCs, enabling its controller to achieve higher bandwidth, whereas the current-measurement path relies on a slower 2 MSPS ADC.

Fig. 8 illustrates the output current, the p-side output voltage, and the p-side output voltage of the HF-LV cell for a 24 A-peak, 64 Hz sinusoidal reference signal. There is a strong correlation between the simulated and measured output-current and output-voltage signals. Nevertheless, some deviations are observed between the measured and simulated p-side output voltage of the HF-LV cell. In the simulation, the HF-LV cell's

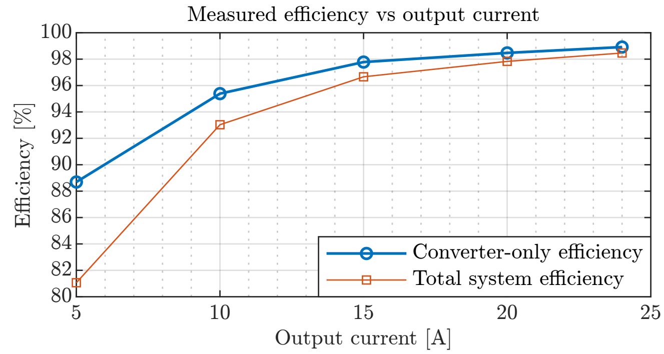


Fig. 10. Measured efficiency as the load current varies from 5 A to 24 A.

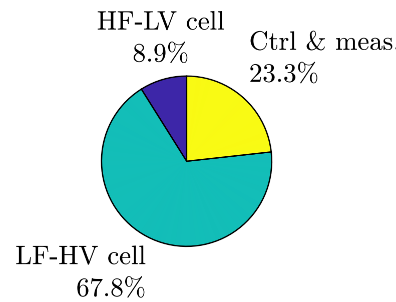


Fig. 11. Loss distribution at 6.45 kW output power (24 A DC load current, 11.2 Ω load). The total loss is 129 W.

output voltage exhibits spikes as it attempts to compensate for dead-time errors when the LF-HV cell transitions between Discontinuous Conduction Mode (DCM) and Continuous Conduction Mode (CCM). However, the simulated model neglects the switching devices' parasitic capacitances. In practice, these capacitances change the switching behavior around the zero-crossing intervals, so the spikes predicted by the simulation do not occur. Furthermore, the HF-LV cell's control effort seems to be larger in the laboratory measurements than in the simulations. Fig. 9 shows the normalized load-current spectrum up to 20 kHz, calculated from the ILA data for a 24 A, 64 Hz sinusoidal reference. The dual-cell converter effectively suppresses baseband harmonics, achieving an impressive Total Harmonic Distortion (THD) of about −100 dB for the first 150 harmonics.

The efficiency of the power amplifier was measured at five DC output current levels (5–24 A), as shown in Fig. 10. The lines connecting the measurement points represent linear interpolation. The efficiency is reported for two cases: 1. Converter-only efficiency, where only the losses of the LF-HV and HF-LV cells are included (shown in blue); and 2. Total system efficiency, where the losses of the power converter, control, and measurement boards are also included (shown in orange). The maximum efficiency is achieved at the maximum load of 24 A. At this point, the power amplifier efficiency is approximately 98.9%, while including the control and measurement losses reduces it to 98.5%. Fig. 11 shows the loss distribution among different parts of the system at the 24 A load. The LF-HV cell accounts for the largest

$$A = \begin{bmatrix} 0 & 0 & -\frac{1}{L_{LF}} & -\frac{1}{L_{LF}} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{L_{HF}} & 0 & 0 \\ \frac{1}{C_{LF}} & 0 & -\frac{R_o + R_{d,LF}}{C_{LF}R_{d,LF}R_o} & -\frac{1}{C_{LF}R_o} & \frac{1}{C_{LF}R_{d,LF}} & 0 \\ \frac{1}{C_{HF}} & \frac{1}{C_{HF}} & -\frac{1}{C_{HF}R_o} & -\frac{R_{d,HF} + R_o}{C_{HF}R_oR_{d,HF}} & 0 & \frac{1}{C_{HF}R_{d,HF}} \\ 0 & 0 & \frac{1}{C_{d,LF}R_{d,LF}} & 0 & -\frac{1}{C_{d,LF}R_{d,LF}} & 0 \\ 0 & 0 & 0 & \frac{1}{C_{d,HF}R_{d,HF}} & 0 & -\frac{1}{C_{d,HF}R_{d,HF}} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L_{LF}} & 0 \\ 0 & \frac{1}{L_{HF}} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$C = \begin{bmatrix} 0 & 0 & \frac{1}{R_o} & \frac{1}{R_o} & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \end{bmatrix}.$$

Fig. 12. State-space representation of the proposed dual-cell power amplifier.

share of the losses, about 68%, followed by the control and measurement boards with approximately 30 W (23%), and finally the HF-LV cell, which contributes only about 11.5 W (9%). As the load current decreases, the contribution of the LF-HV cell to the total losses also decreases, while the losses of the other parts remain nearly constant. Consequently, the overall efficiency drops at lower load levels.

It is worth noting that in high-precision motion systems, the current reference trajectory is generally periodic and exhibits minimal variation, while the load impedance remains nearly constant. This allows the converter to be optimized for a specific operating trajectory. In particular, with the proposed dual-cell topology, the LF-HV cell can be optimized for maximizing efficiency and power density, whereas the HF-LV cell is designed to meet the control bandwidth requirements.

Moreover, the losses in the control and measurement section can be reduced. For simplicity, most ADCs and digital isolators in this design were identical and employed fast LVDS interfaces, which generally consume more power than slower single-ended voltage-based interfaces. However, in practice, only the output current (or voltage) signal requires high-speed measurement. Therefore, by using slower, lower-power ADCs, isolators, and smaller FPGAs, the control-related losses could be reduced by up to one third compared to the current design.

V. Conclusions

This work shows that a dual-cell hybrid power amplifier, driven by a bandwidth-maximizing, decoupled-control strategy, can deliver both ultra-high small-signal bandwidth and high power. The implemented prototype achieves a 1.5 µs current rise time (10%–90%) and settles within 90 µs for small-signal current steps. When tracking a 64 Hz, 24 A-peak sinusoidal reference, the THD remains below -100 dB over the first 150 harmonics, confirming excellent linearity. The system achieves a peak efficiency of 98.5% at a 6.45 kW load. These results validate the feasibility of the proposed architecture and its suitability for precision applications such as high-accuracy motion-system actuators and high-precision audio amplification.

Appendix A
State-Space Model Details

Fig. 12 shows the SSA matrices of the power amplifier.

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